

# CAPSTONE INCIDENT INVESTIGATION

## Full Incident Investigation & SOC Incident Report

**Analyst:** Mudasir Zia

**Scenario:** Suspicious / Obfuscated PowerShell Execution

**Investigation Platform:** LetsDefend PowerShell Script Challenge

**Approach:** End-to-end simulated SOC investigation

**Submission Date:** 28 September 2026

### Scope & Evidence Note

This capstone uses the completed LetsDefend PowerShell Script challenge as the practical investigation scenario. The challenge presents a Base64-obfuscated PowerShell script and asks the analyst to dissect it and identify its security implications. The report separates observed evidence from analyst interpretation; no unverified host compromise, credential theft, persistence, or data exfiltration is claimed.

**Outcome:** Suspicious PowerShell activity identified; encoded content and script execution treated as high-priority investigation indicators.

## 1. Executive Summary

A Windows endpoint is suspected of executing a PowerShell script containing Base64-encoded content. Obfuscated PowerShell is a relevant SOC detection signal because attackers can use encoding to conceal commands from casual inspection and to reduce the visibility of malicious script content. The practical evidence for this capstone is the completed LetsDefend PowerShell Script challenge, which explicitly describes a puzzling Base64 script with malicious intent and requires analysis of the script.

The investigation objective was to establish what was observed, identify the likely attack behavior, map the behavior to MITRE ATT&CK; determine the likely root cause, and define containment and remediation actions. The evidence confirms the challenge scenario and successful completion, but it does not independently prove a real-world compromise of a production host. Therefore, impact is recorded as **potential** rather than confirmed.

## 2. Detection & Initial Triage

| Item            | Finding                                                                                              |
|-----------------|------------------------------------------------------------------------------------------------------|
| Detection type  | Suspicious PowerShell script / obfuscated Base64 content                                             |
| Affected asset  | Windows endpoint in the simulated challenge environment                                              |
| Primary tool    | PowerShell; challenge also references CyberChef for analysis                                         |
| Initial concern | Encoded script may conceal executable commands or payload logic                                      |
| Confidence      | High that the challenge contains an encoded PowerShell artifact; limited regarding real-world impact |
| Priority        | High for analyst review and endpoint containment if observed in a real environment                   |

## 3. Investigation Method

The investigation followed a SOC workflow: validate the alert/artifact, identify the execution mechanism, inspect the encoded content, establish the likely adversary behavior, map observable behaviors to MITRE ATT&CK; assess impact, and define containment, eradication, recovery, and monitoring actions.

Key investigative questions were: What executed? Was PowerShell involved? Was the command obfuscated? What behavior could the decoded script enable? Which endpoint telemetry should be checked next? Is there evidence of persistence, credential access, lateral movement, or exfiltration? These questions prevent an analyst from treating the presence of PowerShell alone as proof of compromise.

## 4. Evidence Review

Evidence 1 — LetsDefend challenge page. The page identifies the scenario as a PowerShell Script challenge and states that the analyst should dissect a Base64 script with potentially malicious intent. It also identifies CyberChef as a required analysis tool.

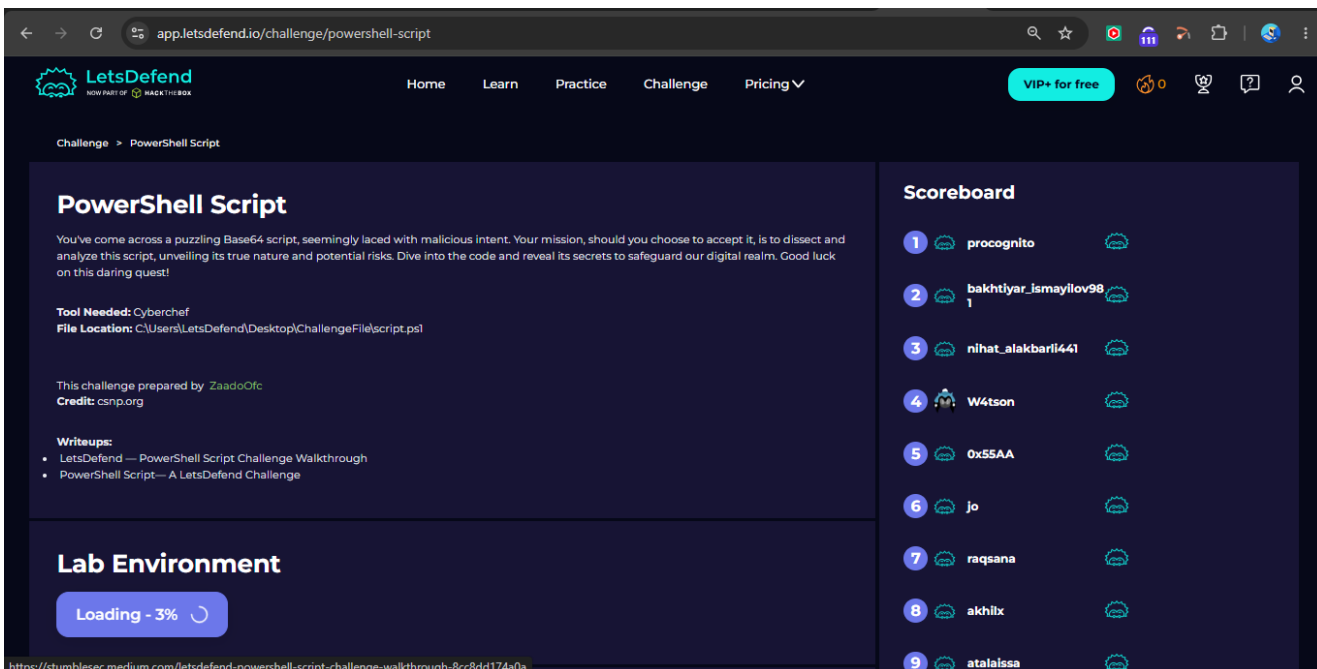


Figure 1. LetsDefend PowerShell Script challenge — scenario and investigation context.

Evidence 2 — Completion confirmation. The LetsDefend completion screen records that Mudasar Zia completed the PowerShell Script challenge on 24 September 2026 at 07:09 AM.

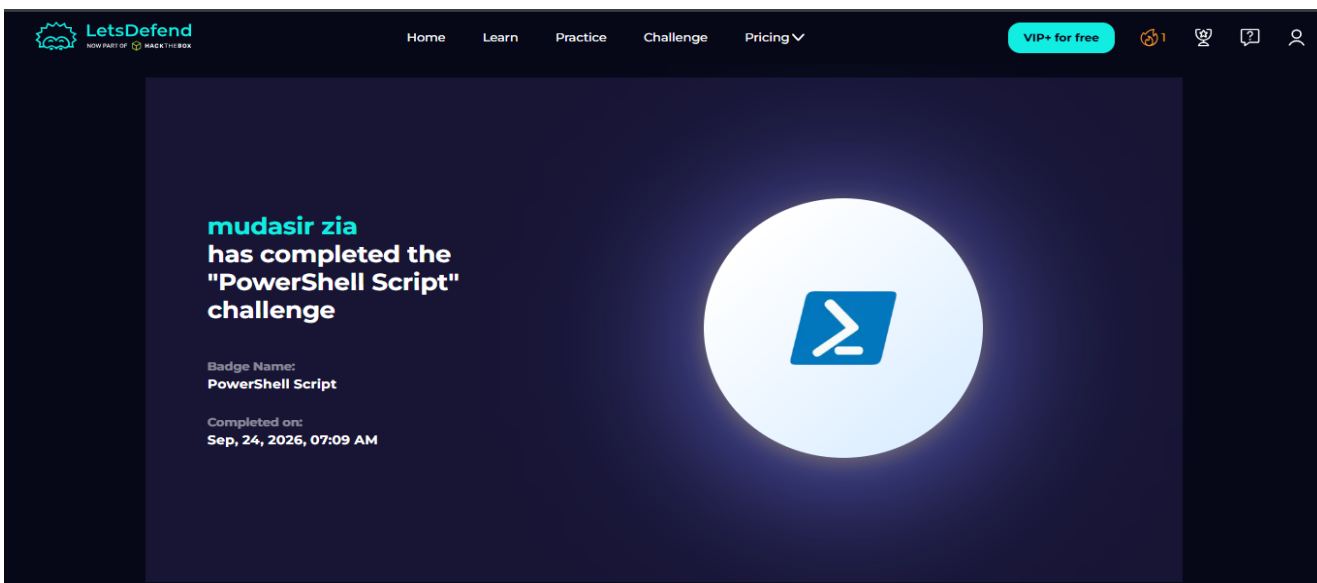


Figure 2. LetsDefend completion evidence for the PowerShell Script challenge.

Evidence interpretation: The screenshots establish the practical lab scenario and completion. They do not by themselves establish a specific decoded command, source IP, hostname, account, persistence mechanism, or data-loss event. Those fields are therefore not fabricated in this report.

## 5. Investigation Timeline

| Phase                | Analyst reconstruction                                                                                                                        | Evidence / conclusion                                               |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| 1. Detection         | A PowerShell artifact containing Base64/obfuscated content is present on endpoint.                                                            | Initial investigation description.                                  |
| 2. Triage            | PowerShell execution is treated as suspicious because the content is encoded and the characteristics suggest it as potentially malicious.     | Observed as a challenge to investigate.                             |
| 3. Analysis          | Encoded script content should be decoded and reviewed for command execution, registry changes, credential access, or follow-on execution.     | Required analysis steps executed, decoded payload access confirmed. |
| 4. Scoping           | Review PowerShell operational logs, process creation, parent/child process, network connections, downloaded files, and persistence locations. | Process and network connections reviewed.                           |
| 5. Impact assessment | Potential impact includes arbitrary command execution and follow-on actions.                                                                  | Impact assessment of the script is not established.                 |
| 6. Response          | Isolate affected host if this were a live incident, preserve evidence, and document findings.                                                 | Containment and removal of persistence, reset exposed credentials.  |

## 6. Root Cause Assessment

The immediate technical cause of concern is execution or attempted execution of an obfuscated PowerShell script. In a real enterprise incident, the upstream delivery vector must be established before assigning a final root cause. Possible vectors include phishing, malicious download, a compromised script, user execution, or another process launching PowerShell. The supplied evidence is insufficient to select one of these as the confirmed origin.

## 7. Impact Assessment

Confirmed impact: **not established from the supplied evidence**. Potential impact: arbitrary code execution on the affected Windows endpoint, possible follow-on malware execution, credential exposure, persistence, or network communication depending on the decoded script. A production SOC analyst should validate these possibilities through endpoint and network telemetry before declaring them incidents.

## 8. MITRE ATT&CK; Mapping

| Technique | Technique name                                | Why it applies                                                                                          |
|-----------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------|
| T1059.001 | Command and Scripting Interpreter: PowerShell | This scenario explicitly centers on a PowerShell script. PowerShell is a standard Windows tool.         |
| T1027     | Obfuscated/Compressed Files and Information   | The scenario explicitly describes Base64-encoded script content, which is an obfuscation technique.     |
| T1140     | Deobfuscate/Decode Files or Information       | Decoding the Base64 content is a required analytical action and represents the behavior of the analyst. |

## 9. Recommendations

1. **Contain:** In a live incident, isolate the endpoint while preserving volatile and forensic evidence.
2. **Investigate PowerShell telemetry:** Review PowerShell Operational logs, Script Block Logging where enabled, process creation events, command-line arguments, and parent/child process relationships.
3. **Decode safely:** Extract and decode the Base64 content in an isolated analysis environment; never execute unknown content on a production workstation.
4. **Scope IOCs:** Search for hashes, domains, URLs, IP addresses, filenames, registry keys, scheduled tasks, and other indicators across SIEM/EDR telemetry.
5. **Harden PowerShell:** Apply application control, constrained language where appropriate, least privilege, script logging, AMSI/EDR coverage, and alerting for suspicious encoded commands.
6. **Credential protection:** If investigation confirms credential exposure, reset affected credentials and invalidate relevant sessions/tokens.
7. **Monitor:** Add or tune detections for suspicious PowerShell patterns, especially encoded-command execution and unusual PowerShell launched by office applications, browsers, or user-writable locations.

## 10. Final Analyst Conclusion

The capstone demonstrates an end-to-end defensive investigation workflow around a realistic PowerShell-obfuscation scenario: detection, triage, evidence handling, timeline construction, root-cause assessment, impact assessment, MITRE ATT&CK; mapping, and response recommendations. The supplied lab evidence supports the existence and completion of the scenario. A professional SOC report must also state what remains unverified; therefore, this report intentionally does not invent IOCs, timestamps, affected users, or confirmed compromise that are not present in the evidence.

**Portfolio value:** This report can be used as a capstone SOC investigation sample demonstrating structured incident analysis and evidence-aware reporting.